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Roll No

CS-4004-CBGS

B.E. IV Semester

Examination, December 2020

Choice Based Grading System (CBGS) **Analysis and Design of Algorithm**

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) What are the rules of manipulate Big-Oh expressions and about the typical growth rates of algorithm?
b) Discuss the steps in mathematical analysis for recursive algorithm. Do the same for finding the factorial of a number.
2. a) Explain divide and conquer technique. Write a recursive algorithm for finding the maximum and minimum element from a list.
b) Discuss Strassen's matrix multiplication and derive its time complexity.
3. Consider the Knapsack instance $n = 3$, $(w_1, w_2, w_3) = (2, 3, 4)$ and $(p_1, p_2, p_3) = (1, 2, 5)$ and $m = 5$. Find the optimal solution.
4. a) Write an algorithm to find single source shortest path? Give an appropriate example also.
b) Explain multi stage graphs with example. Write multistage graph algorithm to forward approach.

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5. Explain the relationship between class P, NP, NP-complete and NP hard problem with example of each class.
6. Consider the following initial cost matrix for Traveling Salesman Problem. Solve this problem using branch and bound technique:

$$\begin{bmatrix} \infty & 4 & 5 \\ 7 & \infty & 5 \\ 4 & 6 & \infty \end{bmatrix}$$

7.
 - a) Apply the Branch and Bound algorithms to solve the travelling salesman problem. Use suitable graph.
 - b) Compare BFS and DFS.
8. Write short notes on any two:
 - a) Master's theorem
 - b) Closest - Pair problem
 - c) B-tree
 - d) Perfect matching in bipartite graphs.
